Benoy Joseph

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EDUCATION

Master of Science, Computer Science **Sep 2025 - May 2027**
Northeastern University *Boston, MA*
- **GPA:** 3.917
- **Coursework:** Database Management Systems, Programming Design Paradigm, Essentials Of Data Science, Algorithms, Machine Learning Operations, Computer Vision And Pattern Recognition

Bachelor of Technology, Computer Science and Engineering **Sep 2020 - May 2024**
Vellore Institute of Technology *Chennai, India*

TECHNICAL KNOWLEDGE

- **Languages and Concepts**: C, C++, Java, Python, R, SQL, Object Oriented Programming
- **Platforms & Databases**: Docker, Kubernetes, Azure, GitHub, MySQL, MongoDB, Oracle, Postgres, AWS
- **Web technologies**: HTML, CSS, JavaScript, React, Node.js, Express, PHP
- **Data Science**: TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn, Data Analytics, Data Visualization
- **Certifications**: Microsoft Certified Azure AI Fundamentals, Full-Stack Web Development Bootcamp

WORK EXPERIENCE

MundoPrints **May 2026 - Aug 2026**
Software Engineer Intern *Boston*
- Built the **LLM-powered instant quote** feature, parsing unstructured customer requests into validated product specifications, constraining the model to the catalog's real attribute schema and scoring extraction and pricing-match confidence to decide whether a quote is auto-confirmed or routed to human review.
- Developed a **RAG chatbot** using OpenAI tool calling over structured catalog and policy data, grounding every answer in retrieved records to eliminate hallucinated products, prices and policy claims.
- Evaluated **extraction accuracy** against real customer phrasing and fixed the main failure modes - unmatched product names, invalid options and lost multi-turn context - through synonym mapping, schema constraints and conversation memory.
- Audited **27 products and 5,972 pricing rules** to surface **459 orderable configurations with no configured price**, then built an **imputation model** deriving each suggested price from existing rules with a confidence score and source rule for analyst review.

HSBC **Jul 2024 - Jul 2025**
Analyst *Chennai, India*
- Supported **10+ automation projects** as part of the **Digital Enablement team**, leveraging Excel, Power BI, Python, Alteryx, and Xceptor to reduce manual effort by **30%** and save approximately **1,000 hours annually**.
- Designed and developed a **React**-based internal marketplace **website** to showcase automation tools using **Git** for version control and collaboration, enabling users to explore different solutions and purchase tools tailored to their workflows, driving a **2×** increase in tool adoption rates.
- Led continuous improvement initiatives by measuring tool utilization, user adoption, and functional coverage across **7+** departments using structured **Excel** logs and **SQL** queries, and built **Power BI** dashboards that identified the top **three** most-used tools and highlighted underutilized solutions, enabling data-driven decisions in monthly stakeholder meetings

PROJECTS

Agent Watch
- Built an end-to-end **ML pipeline** that diagnoses why **LLM** coding agents slow down, attributing each delayed step to a **CPU**, **GPU** **inference**, or **retrieval** bottleneck. It does this by matching agent-step traces with resource telemetry from **OpenTelemetry**, **Prometheus**, and **DCGM** on **GKE**.
- Created a labeled dataset from **50** coding tasks by replaying each one under clean and deliberately overloaded conditions, labeling a step only when its slowdown exceeded a **3σ** noise threshold over the clean baseline. Datasets were versioned in **GCS** and validated with **TFDV**.
- Trained **LightGBM** classification and regression models to predict which resource caused a slowdown and how much of the step's time was avoidable delay, achieving a **10-percentage-point** improvement over a rule-based baseline on mixed-contention classification.
- Deployed the models with calibrated confidence scores that flag low-evidence cases instead of forcing a diagnosis. Added data drift monitoring, automated **Airflow** retraining, and **MLflow**-gated model promotion with rollback, achieving **95%** clean-run specificity and attribution within **~15 s** of step completion.

Research paper on Brain Tumor Classification (Silico Research In Biomedicine Journal Publication)
- Designed a **multiscale CNN-LSTM** with **Discrete Wavelet Transform (DWT)** for automated brain tumor classification, enhancing feature extraction through spatial mode separation.
- Achieved **98.5%** classification accuracy on **2,000 MRI** scans, validated through **5-Fold Cross-Validation** and **Hold-Out** methods to ensure model robustness.

Achievements

- Winner of 2023 WOW Hackathon by Google Developer Student Club (GDSC)
- Best Speaker at HSBC Toastmasters Club (2025)